

A background image showing a hand holding a bone over a dog, illustrating the concept of reinforcement learning. The hand is positioned at the top left, holding a small, light-colored bone. A dog, likely a Jack Russell Terrier, is lying down on the right side, looking up at the bone. The dog has white fur with brown patches on its head and ears, and is wearing a blue collar with a red stripe.

CS 5/7320

Artificial Intelligence

Reinforcement Learning

AIMA Chapter 17+22

Slides by Michael Hahsler
with figures from the AIMA textbook.



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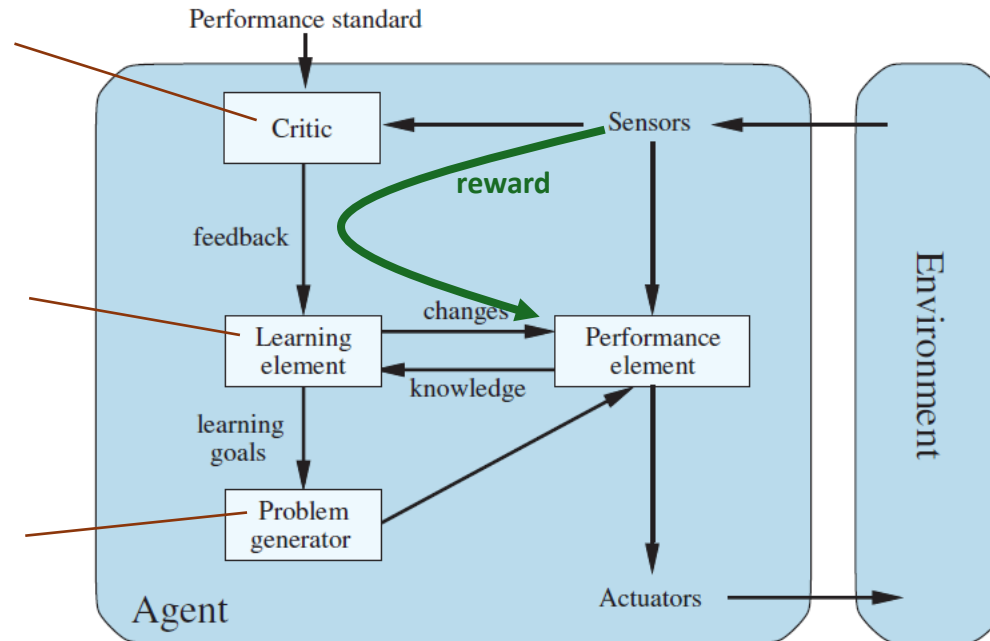
Online Material

From Chapter 2: Agents That Learn

Critic: How is the agent currently performing?

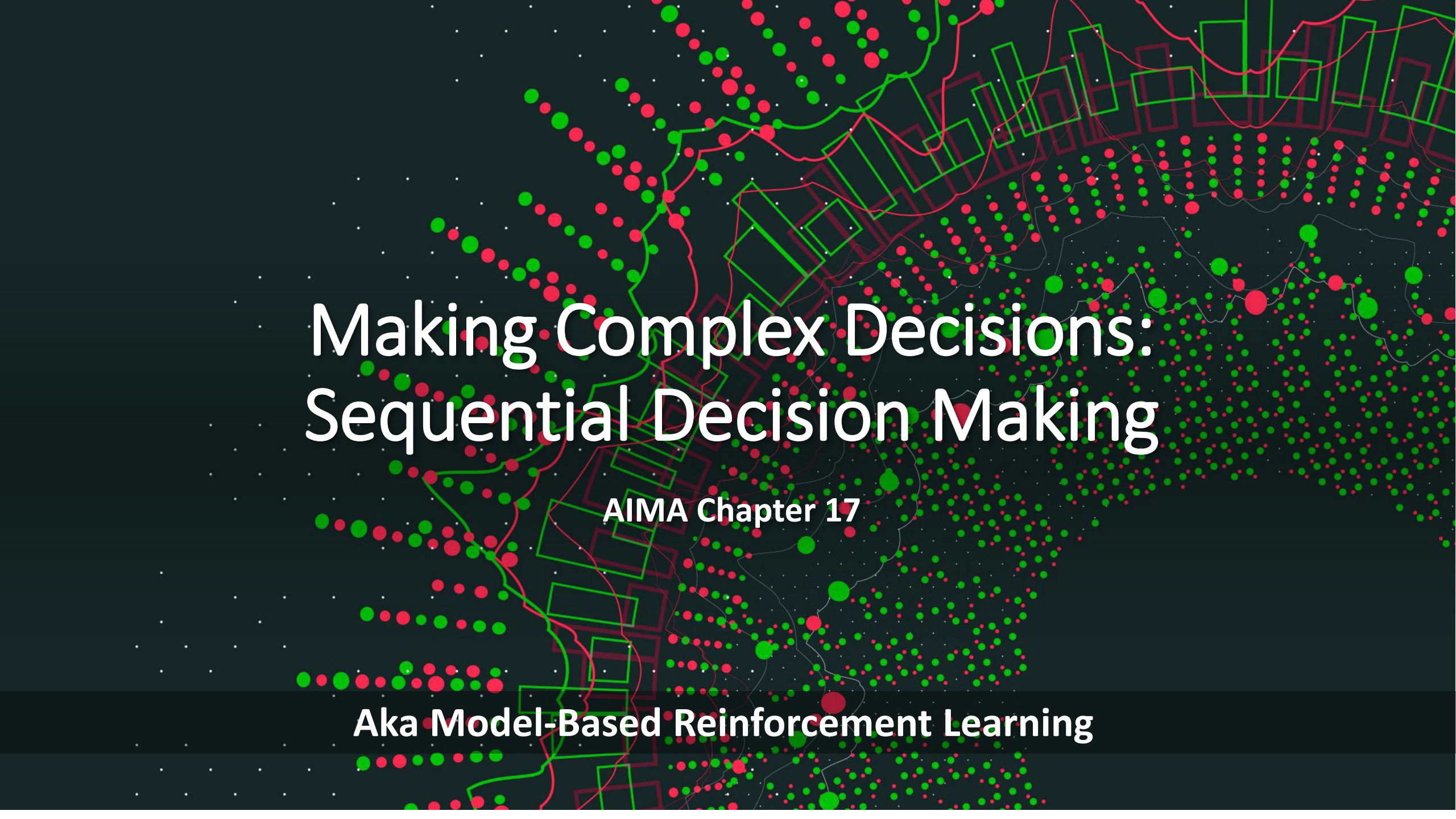
Learning Element: Improves the performance element and changes how it selects actions.
E.g., adding rules, changing weights

Problem generators: Explore new actions.



Positive feedback from the critic, called “reward,” reinforces the performance element.

Reinforcement Learning: How do we learn a good performance element from rewards using trial-and-error?

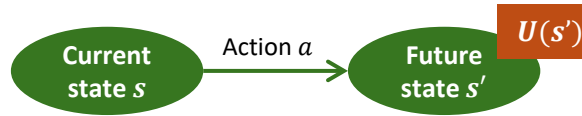


Making Complex Decisions: Sequential Decision Making

AIMA Chapter 17

Aka Model-Based Reinforcement Learning

Remember Chapter 16: Making Simple Decisions



For a decision that we make frequently and making it once does not affect the future decisions (**episodic environment**), we can use the **Principle of Maximum Expected Utility (MEU)**.

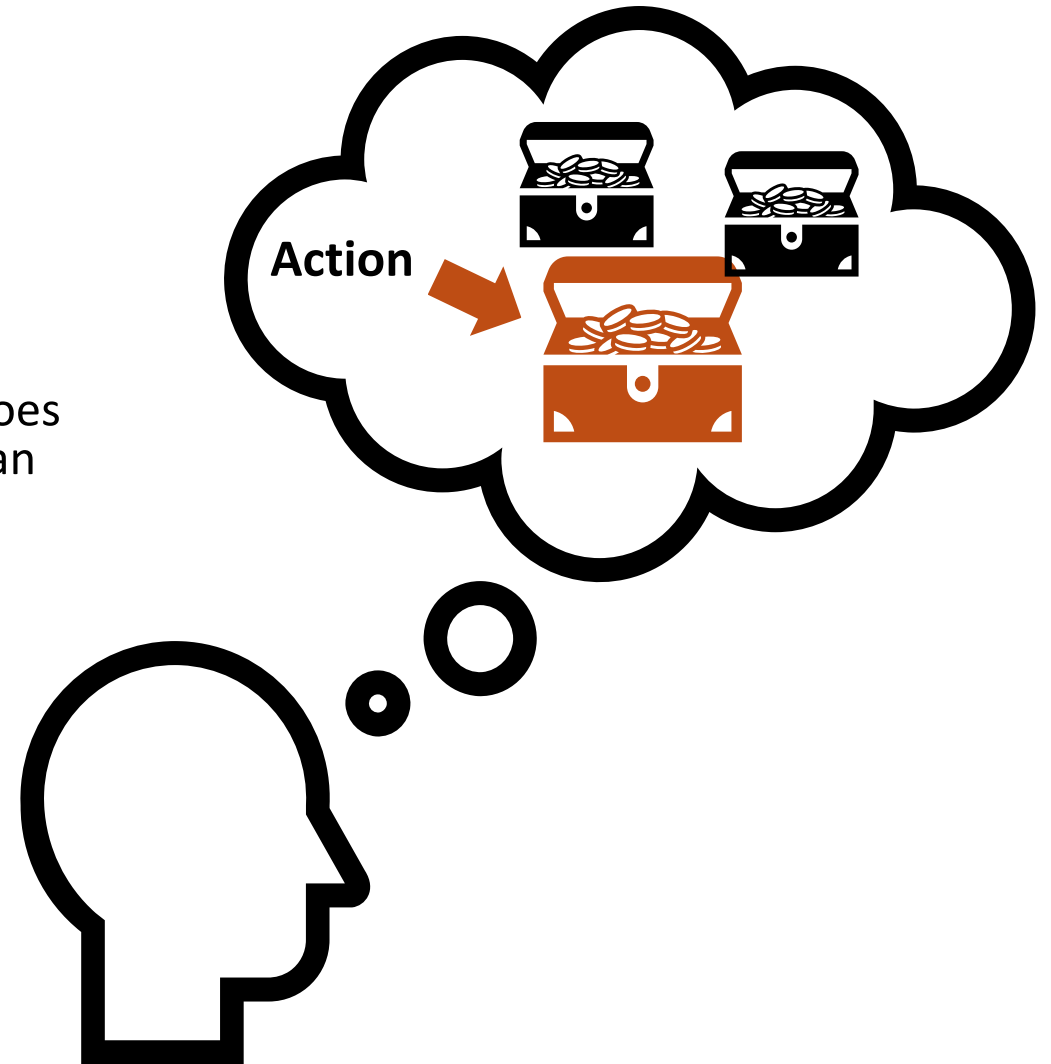
Given the expected utility of an action

$$EU(a) = \sum_{s'} \sum_s P(s) P(s'|s, a) U(s')$$

choose action that maximizes the expected utility:

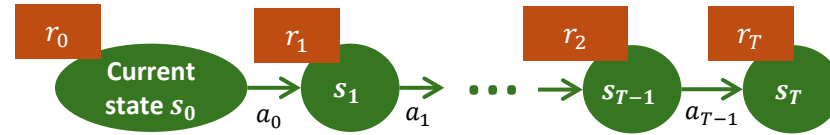
$$a^* = \operatorname{argmax}_a EU(a)$$

Now we will talk about **sequential decision making**.

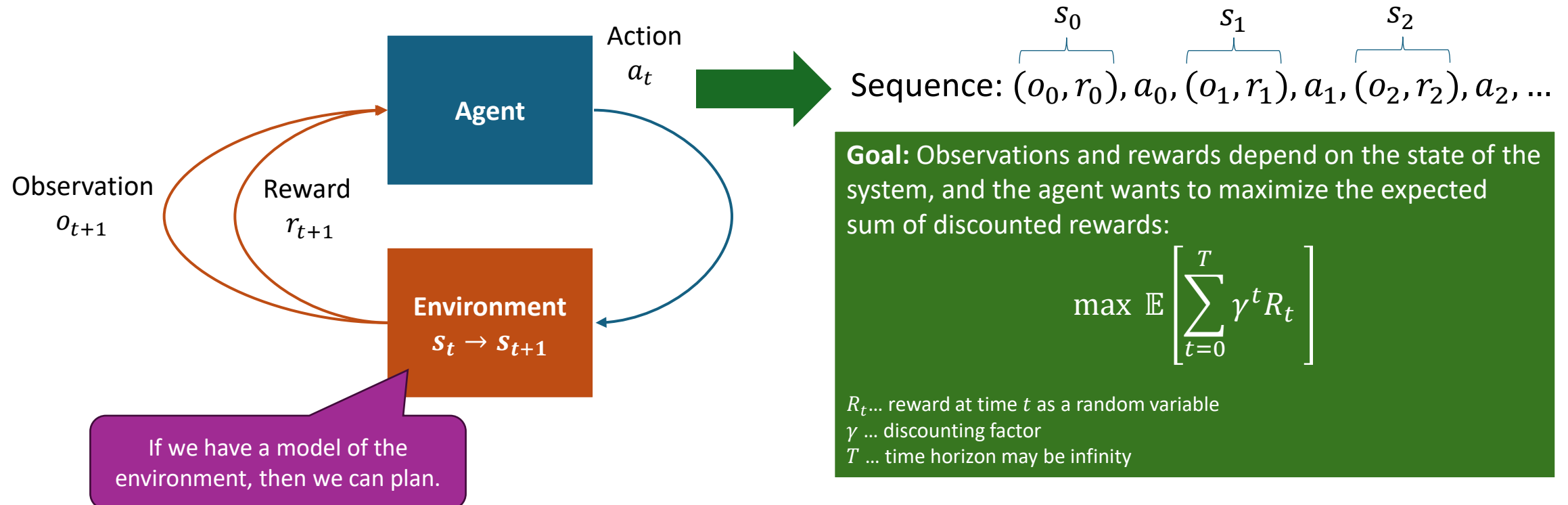


$P(s)$... Uncertainty about current state (= partial observability)
 $P(s'|s, a)$... Stochastic transition function (= non deterministic actions).
 $U(s')$... cardinal utility function.

Sequential Decision Problems



- **Utility-based agent:** The agent's utility depends on a sequence of decisions that depend on each other.
- Sequential decision problems incorporate utility (called immediate and long-term reward), uncertainty, and sensing.



An Environment Model: Markov Decision Process (MDP)

MDPs are discrete-time stochastic control processes defined by:

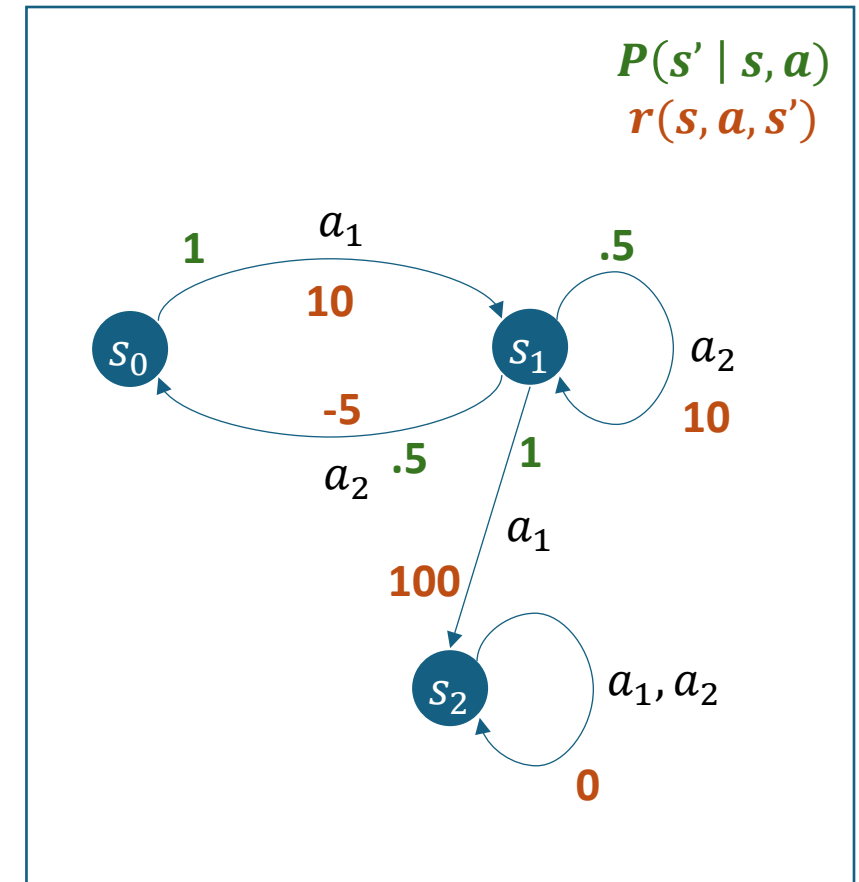
- a finite set of **states** $\mathcal{S} = \{s_0, s_1, s_2, \dots\}$ (initial state s_0)
- a set of available **actions** $ACTIONS(s)$ in each state s
- a **transition model** $P(s' | s, a)$ where $a \in ACTIONS(s)$
- a **reward function** $r(s)$ where the immediate reward depends on the current state (often $r(s, a, s')$ is used to make modelling easier)

MDPs model sequential decision problems with

- a fully observable, stochastic, and known environment;
- a Markovian transition model (i.e., future states do not depend on past states given the current state);
- additive immediate rewards.

Time horizon

- **Infinite horizon:** non-episodic (continuous) tasks with no terminal state.
- **Finite horizon:** episodic tasks. Episode ends after a number of periods or when a terminal state is reached. Episodes contain a sequence of several actions that affect each other.



Example: 4x3 Grid World

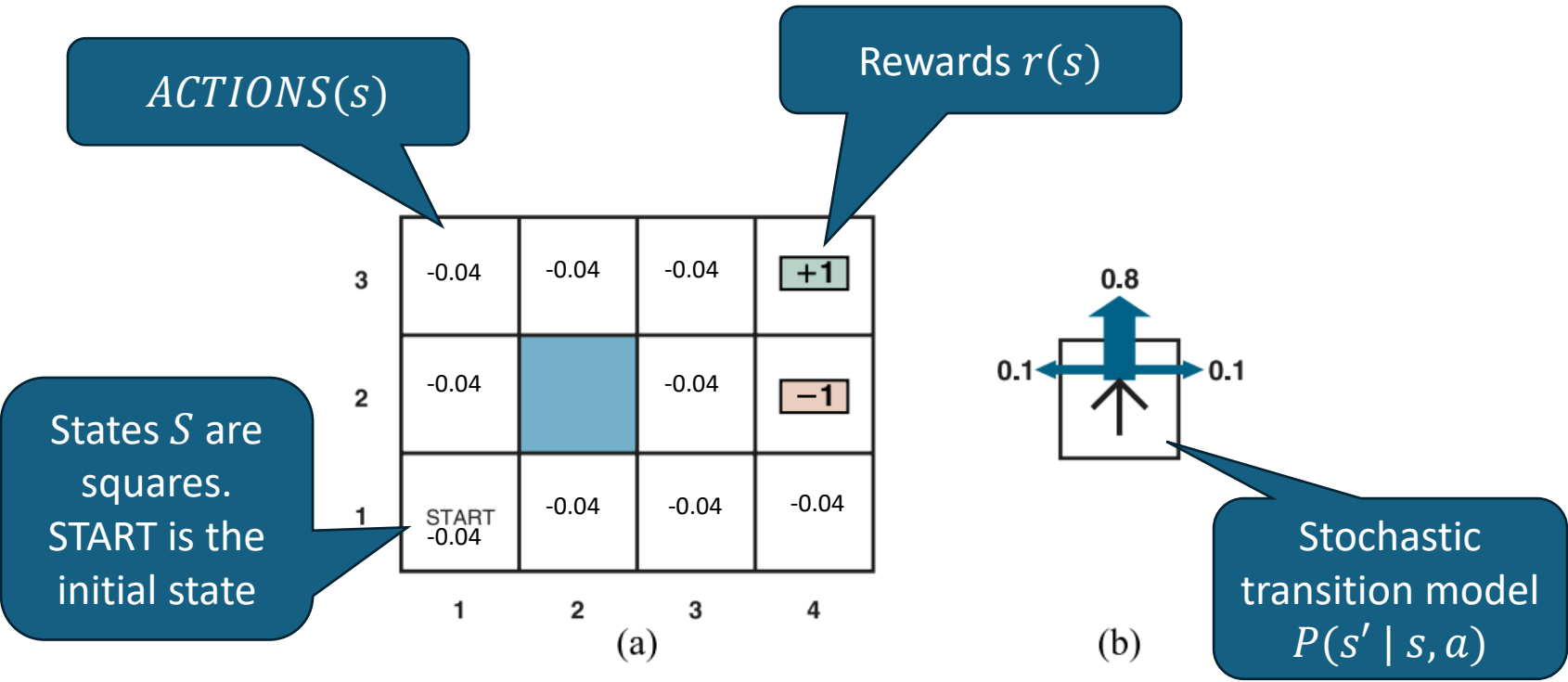


Figure 17.1 (a) A simple, stochastic 4×3 environment that presents the agent with a sequential decision problem. (b) Illustration of the transition model of the environment: the “intended” outcome occurs with probability 0.8, but with probability 0.2 the agent moves at right angles to the intended direction. A collision with a wall results in no movement. Transitions into the two terminal states have reward +1 and -1, respectively, and all other transitions have a reward of -0.04.

Since we know the complete MDP model, we can solve this as a **planning problem**.

For each square: specify what direction should we try to go to maximize the expected total utility.

This is called a **policy** written as the function

$$\pi: S \rightarrow ACTIONS(S)$$

Policy as a Table

s	Action $\pi(s)$
(1,1)	Up
...	...
...	...

Value Function

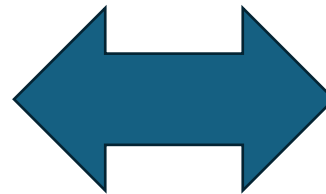
- A policy $\pi = \{\pi(s_0), \pi(s_1), \dots\}$ defines for each state which action to take.
- The expected utility of being in state s under policy π (i.e., following the policy starting from s) can be calculated as the sum of the immediate rewards over the visited sequence of states:

$$U^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t) \mid s_0 = s \right]$$

- $U^\pi(s)$ (often written as $V(s)$) is called **the value function**. It is stored as a table.

Value Function

3	0.8516	0.9078	0.9578	+1
2	0.8016		0.7003	-1
1	0.7453	0.6953	0.6514	0.4279
	1	2	3	4



Value Function

s	State Value $V(s)$
(1,1)	0.7453
(1,2)	0.8016
...	...

γ .. Discounting factor to give more weight to immediate rewards.

$\mathbb{E}_\pi$... Expectation over sequences that can be created by following π .

$r(s)$.. Reward function.

Planning: Finding the Optimal Policy

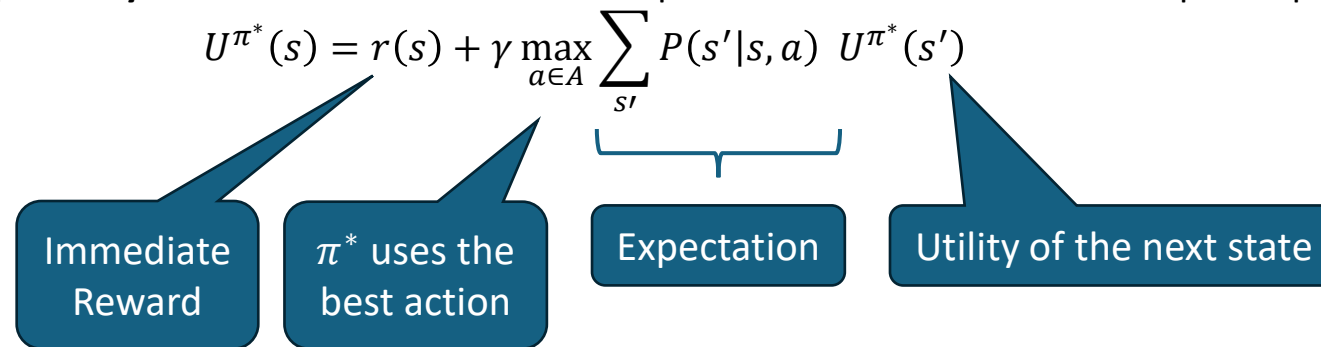
- The goal of solving an MDP is to find an optimal policy π that maximizes the expected future utility for each state

$$\pi^*(s) = \operatorname{argmax}_{\pi} U^{\pi}(s) \quad \text{for all } s \in \mathcal{S}$$

- Issues:**

- a) Optimizing over long sequences is hard: $U^{\pi}(s) = \mathbb{E}_{\pi}[\sum_{t=0}^{\infty} \gamma^t r(s_t) | s_0 = s]$
- b) π^* depends on U^{π} and vice versa!

- Problem a) can be addressed by reformulating the problem recursively as a state consistency problem. The **Bellman optimality condition** states the relationship between state values for the optimal policy:

$$U^{\pi^*}(s) = r(s) + \gamma \max_{a \in A} \underbrace{\sum_{s'} P(s'|s, a)}_{\text{Expectation}} U^{\pi^*}(s')$$


Value Function

3	0.8516	0.9078	0.9578	+1
2	0.8016		0.7003	-1
1	0.7453	0.6953	0.6514	0.4279
	1	2	3	4

Problem b) needs an algorithmic solution.

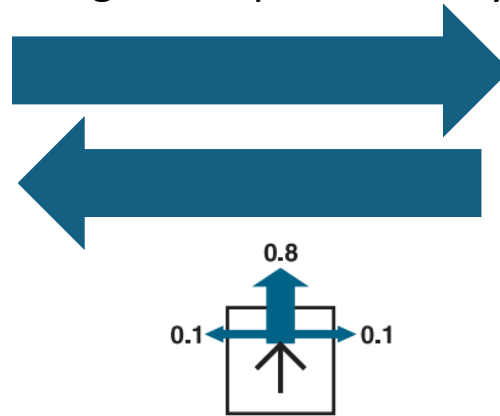
Example Solution: 4x3 Grid World

Value of being in a state $U^{\pi^*}(s)$
(given that we will follow π^*)

3	0.8516	0.9078	0.9578	+1
2	0.8016		0.7003	-1
1	0.7453	0.6953	0.6514	0.4279
	1	2	3	4

$$\gamma = 1$$

Greedy policy:
Always pick the action
leading to the state with
the highest expected utility.



**Optimal action in each state
(policy π^*)**

3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←
	1	2	3	4

It is optimal to walk away from the +1 square to avoid the -1 square!

How do we find the optimal value function/optimal policy?

Value Iteration

Policy Iteration

Q-Function

- $Q(s, a)$ is called the state-action value function. It gives the expected utility of taking action a in state s and then following the policy.

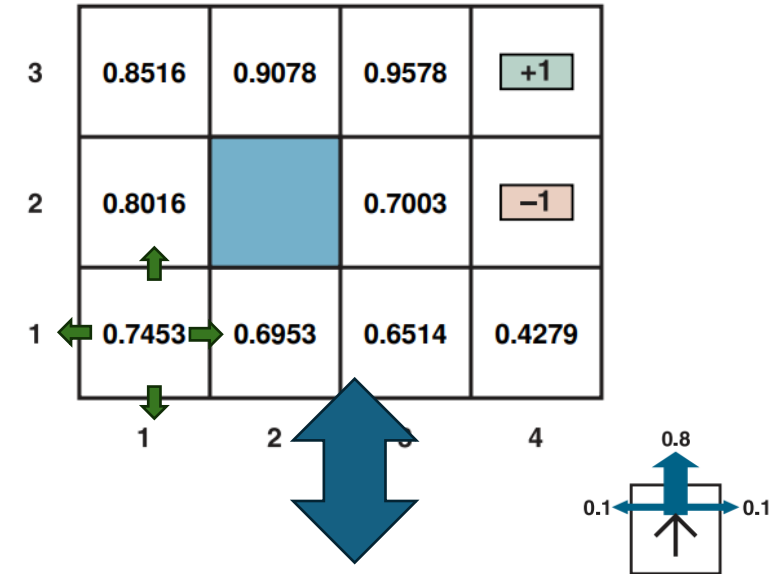
$$Q(s, a) = r(s) + \gamma \sum_{s'} P(s'|s, a) U(s')$$

Immediate
Reward

Expected utility of the
next state

- The Relationship with the state value function: $U(s) = \max_{a \in A(s)} Q(s, a)$
- The Q-function lets us compare the value of taking different action in a given state. It is used in algorithms to determine what action is the best.

Value Function

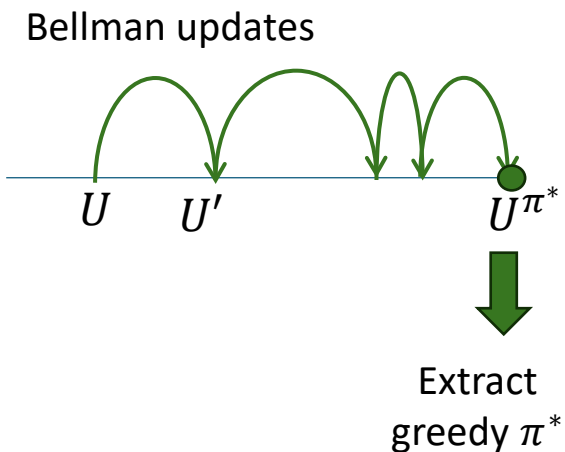


Q-Table

s	a	$Q(s, a)$
(1,1)	Up	0.7453
(1,1)	Right	0.6709
(1,1)	Down	0.7003
(1,1)	Left	0.7109
...	...	...

Value Iteration: Estimate the Optimal Value Function U^{π^*}

Algorithm: Start with a U table (vector) of 0 for all states and then apply the Bellman update over the entries of the table until it converges to the unique optimal solution U^{π^*} .



Guarantee: It will converge because the optimal solution is a fixed point of the Bellman operator.

```
function VALUE-ITERATION( $mdp, \epsilon$ ) returns a utility function
inputs:  $mdp$ , an MDP with states  $S$ , actions  $A(s)$ , transition model  $P(s' | s, a)$ ,
          rewards  $R(s, a, s')$ , discount  $\gamma$ 
           $\epsilon$ , the maximum error allowed in the utility of any state
local variables:  $U, U'$ , vectors of utilities for states in  $S$ , initially zero
                    $\delta$ , the maximum relative change in the utility of any state
```

repeat

$U \leftarrow U'; \delta \leftarrow 0$

for each state s **in** S **do**

$U'[s] \leftarrow \max_{a \in A(s)} \text{Q-VALUE}(mdp, s, a, U)$

if $|U'[s] - U[s]| > \delta$ **then** $\delta \leftarrow |U'[s] - U[s]|$

until $\delta \leq \epsilon(1 - \gamma)/\gamma$

return U

Sweep over
the U table

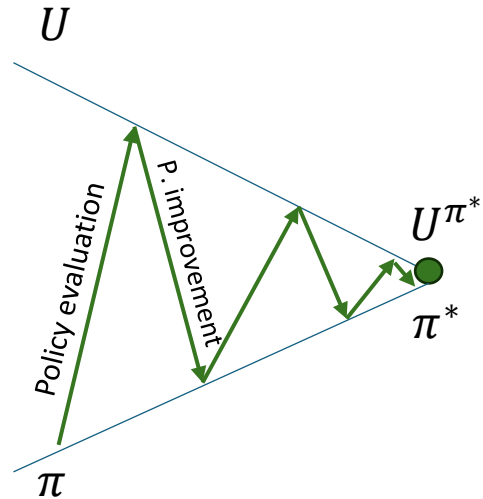
Bellman update: Value of
the best action in state s .

Convergence? Uses a proxy for policy loss
 $\|U^{\pi} - U\|_{\infty}$ as the stopping criterion

U converges to U^{π^*}
and we can extract π^*

Policy Iteration: Find the Optimal Policy π^*

Policy iteration tries to directly find the optimal policy by iterating policy evaluation and improvement.



Guarantee: It will converge because each step improves the utility/policy and there is a finite number of steps.

Greedy policy Improvement.

```
function POLICY-ITERATION(mdp) returns a policy
inputs: mdp, an MDP with states  $S$ , actions  $A(s)$ , transition model  $P(s' | s, a)$ 
local variables:  $U$ , a vector of utilities for states in  $S$ , initially zero
 $\pi$ , a policy vector indexed by state, initially random

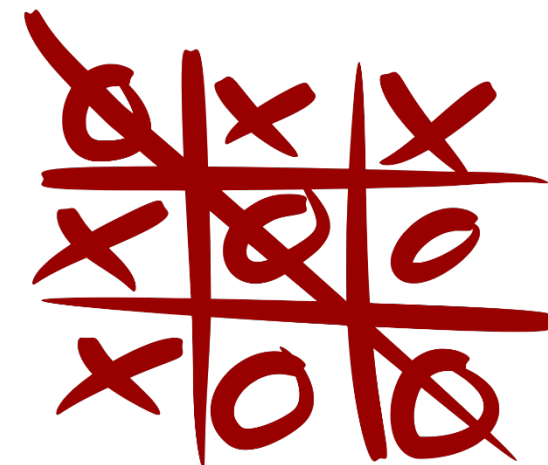
repeat
   $U \leftarrow \text{POLICY-EVALUATION}(\pi, U, \text{mdp})$ 
  unchanged?  $\leftarrow$  true
  for each state  $s$  in  $S$  do
     $a^* \leftarrow \underset{a \in A(s)}{\text{argmax}} \text{Q-VALUE}(\text{mdp}, s, a, U)$ 
    if  $\text{Q-VALUE}(\text{mdp}, s, a^*, U) > \text{Q-VALUE}(\text{mdp}, s, \pi[s], U)$  then
       $\pi[s] \leftarrow a^*$ ; unchanged?  $\leftarrow$  false
  until unchanged?
return  $\pi$ 
```

Estimate U given the current policy (either solve an LP or value iteration with fixed policy)

Convergence test

π converges to π^*
(and U converges to U^{π^*})

Find the Optimal Policy for Tic-Tac-Toe



Definitions from Chapter 5 on Games for a goal-based agent:

s_0	Empty board.	
$Actions(s)$	Play empty squares.	
$Result(s, a)$	Symbol (x/o) is placed on empty square.	→ Stochastic transition model $P(s' s, a)$
$Terminal(s)$	Did a player win or is the game a draw?	
$Utility(s)$	+1 if x wins, -1 if o wins and 0 for a draw. Utility is only defined for terminal states.	→ Reward function $r(s)$

Implementation as a planning model-based RL agent:

1. Model the environment as an MDP. It is completely described by the rules of the game.
2. Plan the optimal policy $\pi^*(s)$ for each state (e.g., using value iteration).
3. Executed the policy.

Potential issues:

- There are many states, so the state value table $U(s)$ has many entries.
- The stochastic transition model $P(s'|s, a)$ needs to be known. We need to assume the other player's policy.
- The tables (value function, policy) are very large. This does not scale for more complicated games (e.g., Connect-4).
- For games, all the rewards are delayed. Immediate rewards are always 0 until the end of the game.

This makes planning hard! An alternative solution is to use online learning with model-free reinforcement learning methods.

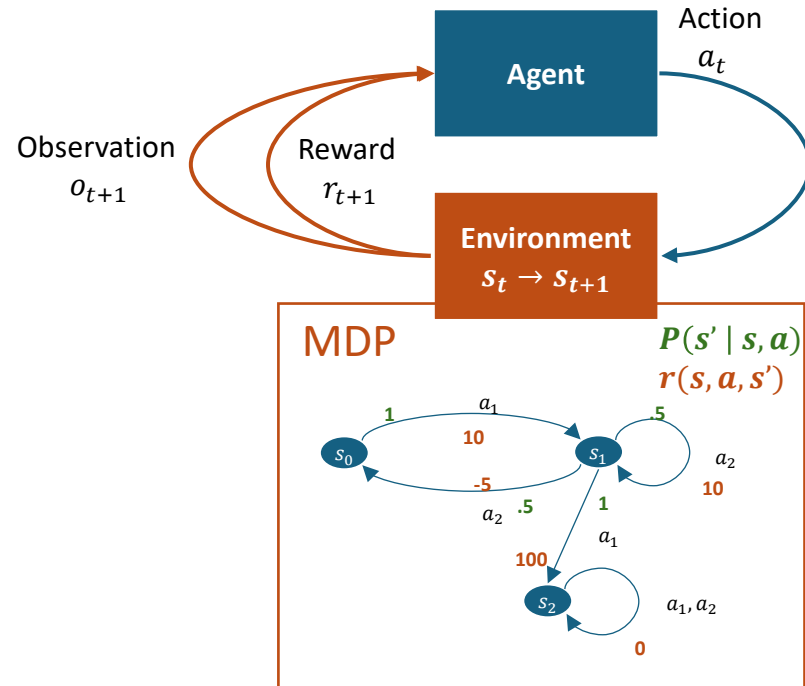


(Model-Free) Reinforcement Learning

AIMA Chapter 22

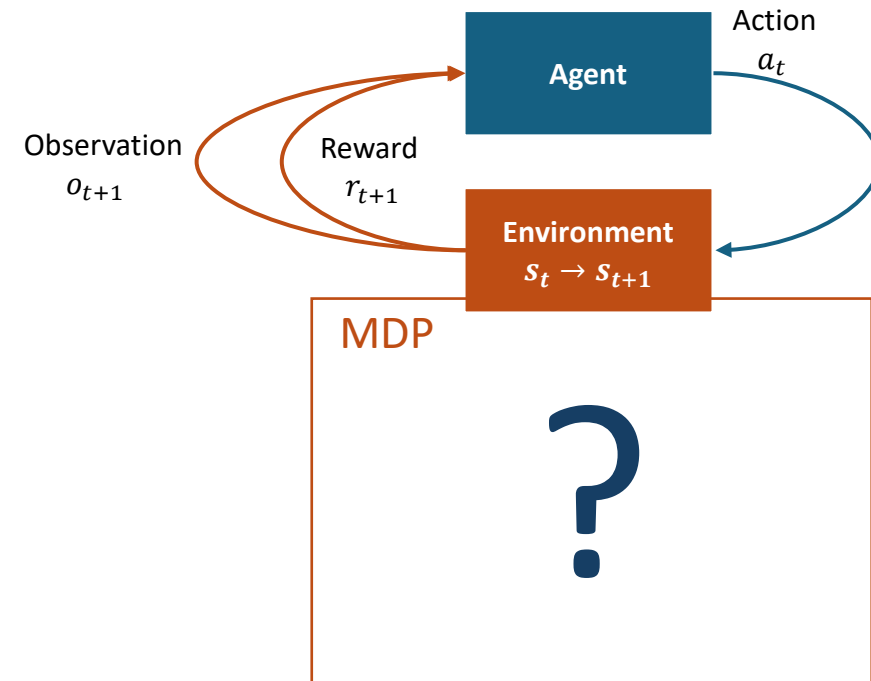
Model-based vs. Model-free Reinforcement Learning (RL)

Model-based RL



Use the MDP model for **planning**
(e.g., value iteration, policy iteration)

Model-free RL



An unknown MDP model means we have
to try actions and use **online learning**.

Reinforcement Learning (RL)

- RL assumes that the problem can be modeled as a **Markov Decision Process (MDP)**.
- However, we do not know the transition or the reward model. This means we have an **unknown environment**, and we need “model-free” methods.
- We cannot use offline planning in unknown environments. The agent needs to explore the environment (try actions) and **use the reward signal to update its estimate of the utility of states and actions**. This is an online learning process that provides positive reinforcement through rewards.
- Key questions are:
 - **Sample efficiency**: How many actions do we need to try in the environment?
 - **Stability**: Does the algorithm converge to a stable solution?
- A popular algorithm is Q-Learning, which tries to learn the state-action value function of important states.

Q-Learning

Q-Learning learns the state-action value function as a table from interactions with the environment.

$$\text{Q-Table} \Rightarrow \pi(s) = \underset{a \in A(s)}{\operatorname{argmax}} Q(s, a)$$

s	a	$Q(s, a)$

function Q-LEARNING-AGENT(*percept*) **returns** an action

inputs: *percept*, a percept indicating the current state s' and reward signal r

persistent: Q , a table of action values indexed by state and action, initially zero

N_{sa} , a table of frequencies for state–action pairs, initially zero

s, a , the previous state and action, initially null

Encodes learned policy

A new episode starts with no previous state.

if s is not null **then**

increment $N_{sa}[s, a]$

$Q[s, a] \leftarrow Q[s, a] + \alpha(N_{sa}[s, a])(r + \gamma \max_{a'} Q[s', a'] - Q[s, a])$

$s, a \leftarrow s', \operatorname{argmax}_{a'} f(Q[s', a'], N_{sa}[s', a'])$

return a

Learning rate

Make $Q[s, a]$ a little more similar to the received reward + the best Q-value of the successor state.

TD-error

Behavior policy: $f(\cdot)$ is the exploration function and decides on the next action. As N increases, it exploits good actions more.

Tabular Methods vs. Value Function Approximation

- U (or Q) tables needs to store and estimate one entry for each state (state/action combination).
- Issues and possible solutions
 - Too many entries to store → lossy compression
 - Many combinations are rarely seen → generalize to unseen entries
- **Idea:** Estimate the state value by learning an approximation function $\hat{U}(s) = h_{\theta}(s)$ based on features of s (ML).
- **Example:** 4x3 Grid World with a linear combination of state features (x, y) and learn θ from observed data.

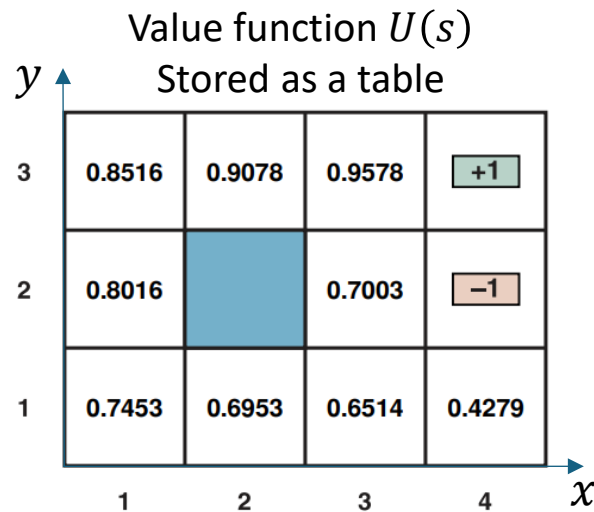


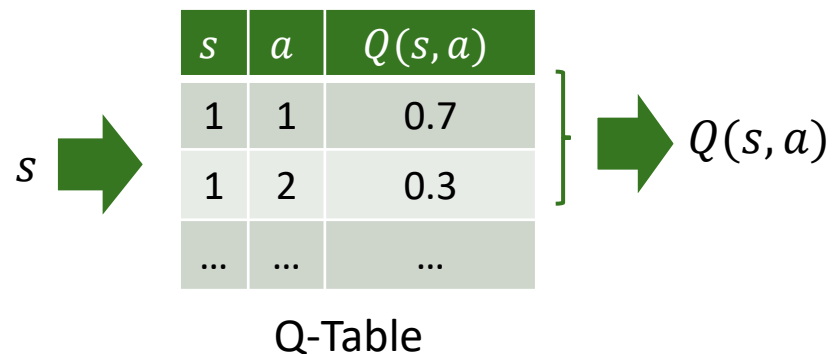
Table vs. approximate $U(s)$

Example: Linear approximation
using state features (x, y)

$$\hat{U}_{\theta}(x, y) = \theta_0 + \theta_1 x + \theta_2 y$$

$\theta = (\theta_0, \theta_1, \theta_2)$ can be updated
iteratively after each new observed
reward using gradient descent.

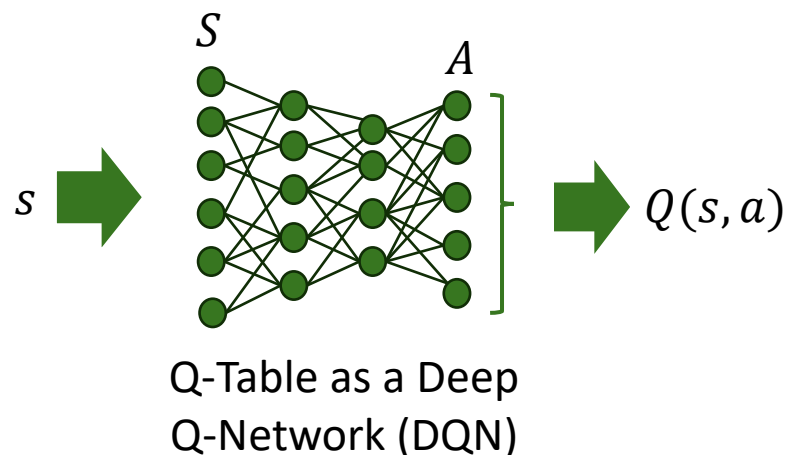
Traditional Tabular Q-Learning



function Q-LEARNING-AGENT(*percept*) **returns** an action
inputs: *percept*, a percept indicating the current state s' and reward signal r
persistent: Q , a table of action values indexed by state and action, initially zero
 N_{sa} , a table of frequencies for state–action pairs, initially zero
 s, a , the previous state and action, initially null

if s is not null **then**
 increment $N_{sa}[s, a]$
 $Q[s, a] \leftarrow Q[s, a] + \alpha(N_{sa}[s, a])(\underbrace{r + \gamma \max_{a'} Q[s', a']}_{\text{target}} - \underbrace{Q[s, a]}_{\text{prediction}})$
 $s, a \leftarrow s', \operatorname{argmax}_{a'} f(Q[s', a'], N_{sa}[s', a'])$
return a

Deep Q-Learning



Target networks: It turns out that the Q-Network is unstable if the same network is used to estimate $Q(s, a)$ and also $Q(s', a')$. Deep Q-Learning uses a second target network for $Q(s', a')$ that is updated with the prediction network every C steps.

Experience replay: To reduce instability more, generate actions using the current network and store the experience $\langle s, a, r, s' \rangle$ in a table. Regularly use samples from the table to update the networks..

Loss function: squared difference between prediction and target.



Summary

- Agents can learn the value of being in a state from **reward signals**.
- Rewards can be delayed (e.g., at the end of a game).
- **Unknown transition models** lead to the need for exploration by trying actions (model-free methods such as Q-Learning).
- All RL problems are computationally very expensive and often can only be solved by **approximation**. The state-of-the-art approach is to use deep artificial neural networks for function approximation.
- Not covered here: Not being able to **fully observe the state** makes the problem more difficult and leads to Partially Observable MDPs.